



# DEBRE BRHAN UNIVERSITY

COLLEGE OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE

Individual Assignment-I

Topic: The Job Assignment Problem Algorithmic Strategies Using Branch and Bound

Name : *Emam lakew*.....*DBU1405606*

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# 1 Algorithm Definition and Conceptual Framework

## 1.1 Problem Definition

The Job Assignment Problem is a fundamental combinatorial optimization challenge. Given a set of  $n$  agents and  $n$  jobs, along with a cost matrix  $C$  where  $C[i][j]$  represents the cost of assigning agent  $i$  to job  $j$ , the objective is to find a bijective mapping (one-to-one) that minimizes the total cost:

$$\text{Minimize } Z = \sum_{i=1}^n \sum_{j=1}^n C_{ij} x_{ij}$$

Subject to the constraints that each agent and each job are paired exactly once.

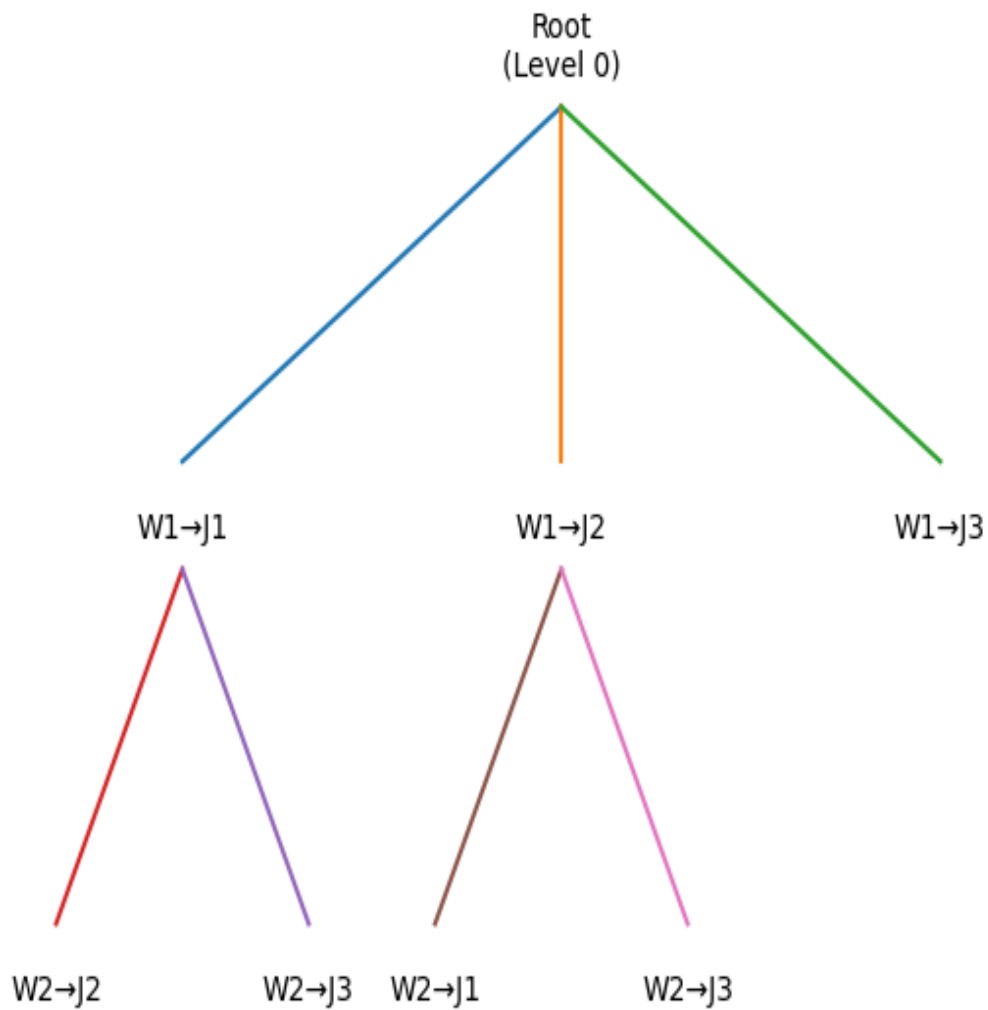
## 1.2 The Principles of Branch and Bound (B&B)

Branch and Bound is an optimization strategy that avoids the  $n!$  complexity of exhaustive search through two primary mechanisms:

- **Branching:** The problem is decomposed into a state-space tree. Each level of the tree represents an assignment for a specific agent.
- **Bounding:** At each node, a *Lower Bound* (LB) is calculated. This is a heuristic estimate representing the best possible outcome if we continue searching from that node.
- **Pruning:** If a node's LB is greater than or equal to the cost of a previously discovered complete solution (the *Upper Bound*), that entire subtree is discarded.

### 1.3 Graphical Representation

The following diagram illustrates the state space tree for the Job Assignment Problem using the Branch and Bound strategy. Each level represents assigning a job to the next worker. Branches that exceed the current best bound would be pruned during execution.



## 2 Pseudo-code Implementation

This implementation uses a Least-Cost Search utilizing a Priority Queue to prioritize nodes with the lowest bounds.

```
Algorithm: JobAssignment_B&B( CostMatrix[n][n])
1. Initialize PriorityQueue PQ
2. Root.path = [], Root.cost = 0, Root.level = 0
3. Root.bound = CalculateInitialBound( CostMatrix)
4. PQ.Push(Root)
5. GlobalMin = Infinity

6. While PQ is not Empty:
    Current = PQ.PopMin()

    If Current.bound >= GlobalMin:
        Continue (Prune Node)

    If Current.level == n:
        GlobalMin = Current.cost
        BestPath = Current.path
        Continue

    For j from 0 to n-1:
        If Job j is not in Current.path:
            Child = CreateNode(Current.level + 1)
            Child.path = Current.path + [j]
            Child.cost = Current.cost + CostMatrix[Current.level][j]
            Child.bound = CalculateBound(Child, CostMatrix)

            If Child.bound < GlobalMin:
                PQ.Push(Child)

7. Return BestPath, GlobalMin
```

Listing 1: Job Assignment via Branch and Bound

## 3 Time and Space Complexity Analysis

### 3.1 Time Complexity

- Worst-Case  $O(n!)$ : Occurs when the cost matrix is "flat" (all costs are similar), forcing the algorithm to visit every possible node.
- Average-Case: Significantly more efficient than brute force. Pruning typically reduces the search space to a small fraction of the total permutations.
- Best-Case  $O(n^2)$ : Occurs when the first path found is the absolute minimum and all other bounds are immediately higher.

### 3.2 Space Complexity

The space complexity is  $O(bd)$ , where  $b$  is the branching factor and  $d$  is depth. Because we store active nodes in a Priority Queue, memory usage can be intensive compared to simple Back-tracking

## 4 Comparative Analysis

### 4.1 Strengths vs. Other Algorithms

- vs. Backtracking: While Backtracking finds the same result, it only prunes when a path becomes invalid. B&B prunes when a path becomes *inefficient*, making it much faster for cost minimization.
- vs. Greedy Algorithms: Greedy approaches are  $O(n^2)$  but often fail to find the global minimum. B&B guarantees optimality.

### 4.2 Weaknesses

Compared to Dynamic Programming (Bitmasking), B&B has an unpredictable runtime. DP has a fixed  $O(n^2 2^n)$  complexity, whereas B&B can vary based on the data distribution.

## 5 Proposed Contribution: Look-Ahead Bounding

A common limitation of B&B is "loose bounds." I propose a Look-Ahead Heuristic. Instead of summing the minimum of the remaining rows independently, the bounding function should check if the "best" job for Agent  $i$  is also the "best" for Agent  $i + 1$ . Impact on Complexity: This adds an  $O(n \log n)$  sorting overhead per node but can reduce the total number of nodes generated by over 50% in dense matrices, leading to a faster overall completion time for large-scale problems.

## 6 Conclusion

The Job Assignment Problem exemplifies a classic combinatorial optimization challenge in algorithm design, where the goal is to assign agents to jobs at minimum total cost. The Branch and Bound (B&B) strategy provides an effective framework to solve this problem optimally by systematically exploring the solution space while pruning suboptimal paths. By leveraging branching, bounding, and pruning mechanisms, B&B significantly reduces the computational effort compared to exhaustive search, while guaranteeing the global minimum solution.

Although B&B can have high memory usage and an unpredictable runtime depending on the cost matrix, it outperforms simple Backtracking and Greedy approaches in both accuracy and efficiency. Proposed enhancements, such as Look-Ahead Bounding, further optimize performance by tightening bounds and reducing the number of nodes generated, particularly for dense or large-scale matrices.

Overall, the study of B&B for the Job Assignment Problem highlights the importance of combining rigorous algorithmic strategies with heuristic improvements to achieve both optimality and practical efficiency in complex optimization tasks.

## **7** References

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2. Levitin, A. (2012). *The Design and Analysis of Algorithms*. Pearson.
3. Horowitz, E., & Sahni, S. (2008). *Fundamentals of Computer Algorithms*. Galgotia Publications.